

600M Events a Day

Tiered storage & serving from Kafka to four clocks

ONE LOG • FOUR SLAS • ZERO FULL SCANS

The ask: one stream, four clocks

SCALE MATH

6,944
avg eps

35K
peak eps

VOLUME

0.9 — 1.2 TB / day raw

1.5KB Avro vs 3KB JSON, ZSTD

RETENTION

2 years = 438 — 650 TB in S3 ZSTD

vs 2.6 PB in Kafka RF3 → 17x cost

PARTITIONS

72 partitions @ ~500 eps each

key=user_id, acks=all, idempotent



PERSONALIZATION • ONLINE

Latest activity in seconds

<3s Flink → ScyllaDB • p95 <15ms

FRAUD • STREAM

Purchase events in minutes

<2m Filter 5% • 5-min window

ANALYTICS • LAKEHOUSE

Dashboards accept 15 min lag

15m StarRocks on Gold • was scanning 2y

TRAINING • BATCH

Next-day features T+1

T+1 Iceberg time-travel • Feast offline

COMPLIANCE • DELETION STAGE

Right-to-erasure fans out to every store

<1h online, <24h lake • crypto-shred + equality deletes

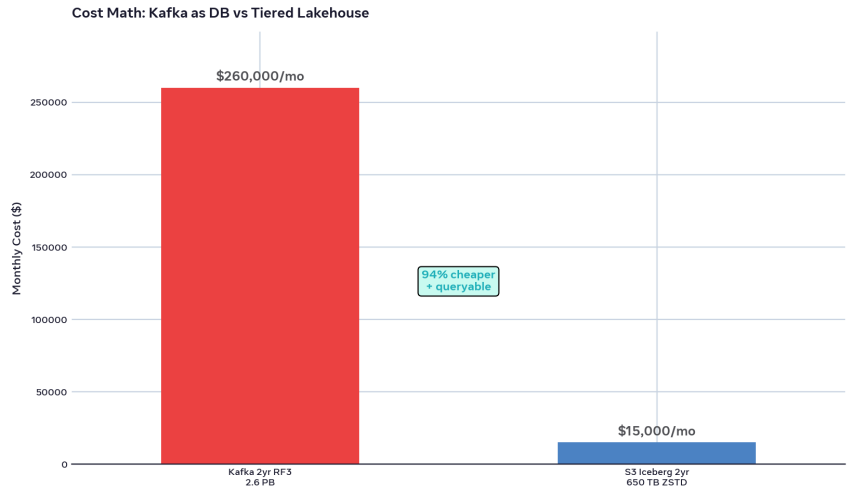


Anti-pattern: Kafka as database

BEFORE • 2YR RF3 IN KAFKA

COST	**2.6 PB in Kafka (1.2 TB × 730 × 3) = $260k/mo** vs 650 TB S3 ZSTD = $15k/mo → 94% more
QUERY	**No partition pruning, no predicate pushdown** WHERE event_date='2026-10-01' scans full 2 years
GDPR	**Log deletion by key impossible** Tombstones don't remove data, compacted topics leak PII
OPS	**7-day hot + Tiered Storage is the fix** Kafka = transport, S3 = truth • replay from Bronze

```
retention.ms=7d
confluent.tier.enable=true
local.retention.ms=604800000
remote.storage.enable=true
```



AFTER • 7-DAY HOT + S3 ICEBERG

Kafka keeps **7 days hot locally**, tiered to S3 via KIP-405. Flink S3 sink builds Bronze. Iceberg Silver gives **days(event_time) hidden partitioning** + Parquet stats.

Result: 650TB → 50GB scanned, 3-5 sec Athena.

3 principles for 600M / day

01 TIER EVERYTHING

Hot → Warm → Cold, not hot forever

Kafka holds 7 days hot, tiered to S3 after. S3 is truth. Small files die by compaction.

- → Kafka 7d local + Tiered S3 remote 2y
- → Bronze raw 128MB / Silver 512MB Parquet ZSTD
- → Manifest pruning = skip 99.99%

02 SEPARATE SPEED & TRUTH

Flink for seconds, lakehouse for 15 min / T+1

No single store can do p95 10ms and scan 650 TB. Split the clocks.

- → Speed: ScyllaDB / Cassandra upsert
- → Truth: Iceberg Silver days(event_time)
- → Serving: StarRocks Gold 15m dbt

03 ERASURE BY DESIGN

Deletion is a stream, not a script

Per-user DEK encrypted PII + equality deletes. One request fans out everywhere.

- → Kafka topic user_deletion_requests
- → Iceberg equality delete files, merge at read
- → KMS shred: instant unreadable

BONUS

Avro + Schema Registry + ZSTD = 60% smaller than JSON — 3KB → 1.5KB, 0.9TB/day not 1.8TB, faster SerDe, evolution safe

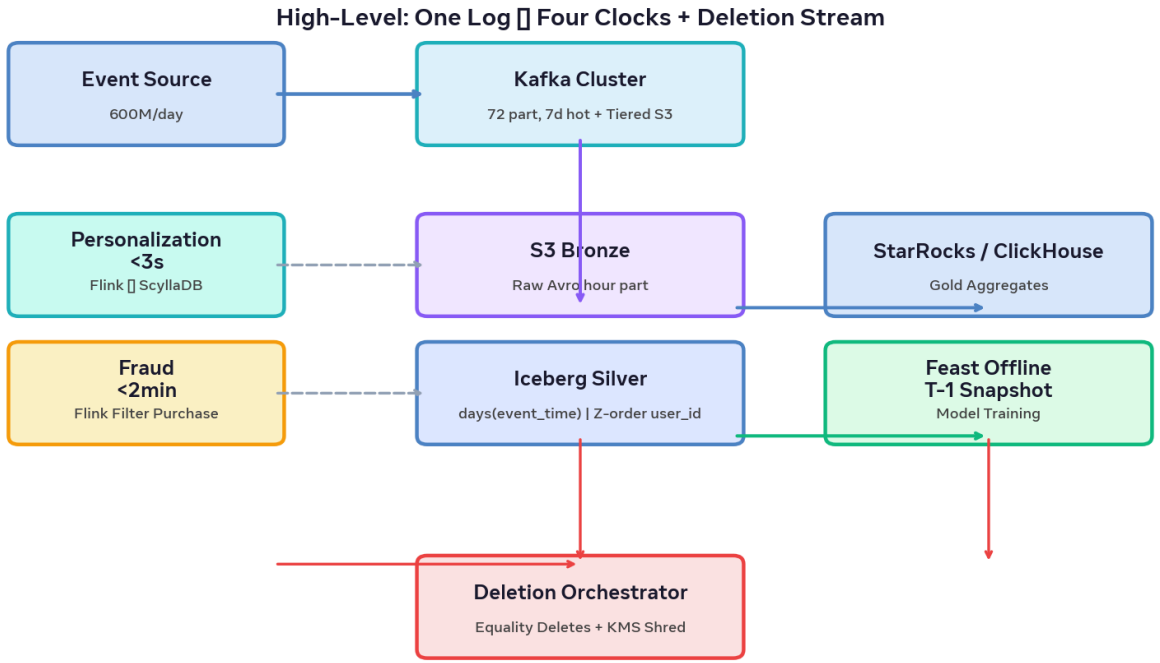
```
event_id uuid / user_id str / type enum /
pii_encrypted bytes
```

End-to-end flow

- One log → four clocks + deletion stream • 72 partitions keyed by user_id

KAFKA → S3 BRONZE → ICEBERG SILVER → SERVING • TIERED 7D HOT + S3

✓ EXACTLY-ONCE FLINK → S3



PATH A • PERSONALIZATION <3S

ScyllaDB online feature store

Kafka → Flink keyBy(user_id) RocksDB last 20 → upsert user_activity

Flink | ScyllaDB p95 10ms

PATH B • FRAUD <2MIN

Filtered purchase stream

WHERE type='purchase' ~30M/day • velocity check • alert

Flink filter | Cassandra audit

PATH C • ANALYTICS 15M

Iceberg Silver + StarRocks Gold

S3 Bronze 60s rolling → days(event_time) → Gold aggregates 15min dbt

Parquet ZSTD 512MB | StarRocks

PATH D • TRAINING T+1

Iceberg snapshot → Feast offline

FOR TIMESTAMP AS OF T-1 → offline store → Spark train

Time travel | Feast

DELETION ORCHESTRATOR

Equality deletes + KMS crypto-shred

user_deletion_requests topic fans out to all 5 stores

Kafka as shock absorber • 72 partitions

Cluster: MSK / Confluent, 3 AZ, r6g.xlarge
72 partitions keyed by user_id % 72, acks=all, idempotent, compression.type=zstd. 35K eps peak is trivial — 500 eps per partition.

Schema: Avro + Registry + field-level encryption
Per-user DEK, KEK in KMS. pii_encrypted bytes. Envelope keeps GDPR shred instant. 1.5KB vs 3KB JSON = 60% smaller.

Tiered Storage: 7d hot, 2y remote
local.retention.ms 7d, tiering to S3. Kafka = transport, S3 = truth. Replays from Bronze, not Kafka log.

Exactly-once: Flink checkpoints to S3
Transactional sink, checkpointing 2-min. Bottleneck is downstream compaction, not ingest.



0.9-1.2 TB/d
RAW INGEST ZSTD

7d → S3
HOT → TIERED

topic: ecommerce.events
partitions=72, key=**user_id**
retention.ms=**7d** + tiered
Avro + **Schema Registry**
DEK per user → **KMS KEK**
compression=**zstd**

User's latest activity in seconds

PATH A • PERSONALIZATION <3S

Kafka → Flink keyBy(user_id) → ScyllaDB

Flink keeps last 20 events per user in RocksDB state. On each event, upsert with TTL 2y. Truth stays in Iceberg Silver.

```
stream.keyBy(e ⇒ e.user_id)
.process(RocksDB Last20)
.addSink(ScyllaUpsert)
checkpoint: 2min S3 exactly-once
```

p95 10ms	p99 25ms	35K
SCYLLA READ	API EDGE	PEAK RPS TRIVIAL

ALTERNATIVE

Redis Cluster <5ms but 5x cost

Scylla gives wide-row, LSM, lower $/GB for 600M writes/day. Use Redis only for top 1% VIP if needed.

SCYLLADB USER_ACTIVITY

PK user_id — LWW

COLUMNS

last_event_ts
last_event_type
last_20_events list<json>
session_id

READ PATH

API → ScyllaDB
GraphQL federated
Cache-Control no-cache
Fan-out: Iceberg source-of-truth



Purchase stream in minutes

Kafka 72p → Flink SQL Filter → Enrich → 5-min window → Dual sink • SLA 2 min checkpoint

1

FILTER • 5% OF STREAM

WHERE type='purchase' → 30M / day

600M × 0.05 = 30M purchases. Flink SQL pushdown, no custom code.

2

ENRICH • ASYNC I/O

Async lookup ScyllaDB velocity

Purchases last 10 min, last device, geo drift. Timeout 20ms, retries bounded.

3

STATEFUL • TUMBLING WINDOW

5-min window keyBy(user_id), count > threshold → Alert

RocksDB state backend, checkpoint interval 2 min. Exactly-once to sinks.

SINK 1 • SYNC — P50 <80MS

Fraud API

POST /v1/fraud/check
velocity features → score

SINK 2 • AUDIT — CASSANDRA

```
CREATE TABLE purchases_by_user (
  user_id text,
  purchase_time timestamp,
  amount decimal,
  PRIMARY KEY (user_id, purchase_time)
) WITH CLUSTERING ORDER BY (purchase_time DESC);
```

THROUGHPUT MATH — 30M/DAY

347	1.7K
avg eps purchases	peak eps purchases

Lakehouse — bronze → silver → gold

FIX FULL SCANS • **15-min freshness, 50 GB scanned not 650 TB**

PARTITION EVOLUTION • HIDDEN DAYS(EVENT_TIME) • Z-ORDER USER_ID • MANIFEST PRUNING

FLINK 60S ROLLING • SPARK REWRITE 512 MB



BRONZE • RAW LANDING

S3 raw Avro hour= part

Flink S3 sink exactly-once, 60s rolling, 128 MB files. Handles late data, replayable from any hour.

```
s3://lake/bronze/
hour=2026-10-13-14/
file-128MB-0001.avro
```

1.2 TB / day | Avro + ZSTD

SILVER • SOURCE OF TRUTH

Iceberg silver.events

Hidden partitioned by days(event_time), sorted by user_id. ZSTD Parquet 512 MB target.

```
PARTITIONED BY days(event_time)
SORTED BY user_id
TBLPROPERTIES (format=parquet, codec=zstd)
manifest pruning • stats pruning
```

Compaction 15m | Time travel T-1

GOLD • SERVED AGGREGATES

dbt → daily_kpi, funnel_hourly

dbt jobs every 15 min materialize into StarRocks / ClickHouse Gold. No dashboard hits Silver directly.

```
gold.daily_kpi
gold.funnel_hourly
gold.feature_agg
→ StarRocks primary key
```

15-min dbt | Sub-sec dashboards

From full scan to 50GB

BEFORE • NO PRUNING

650 TB scanned

WHERE event_date='2026-10-01' scans 2 years, no partition.

AFTER • ICEBERG PRUNING

50 GB scanned

days(event_time) hidden + min/max stats → 99.99% pruned.

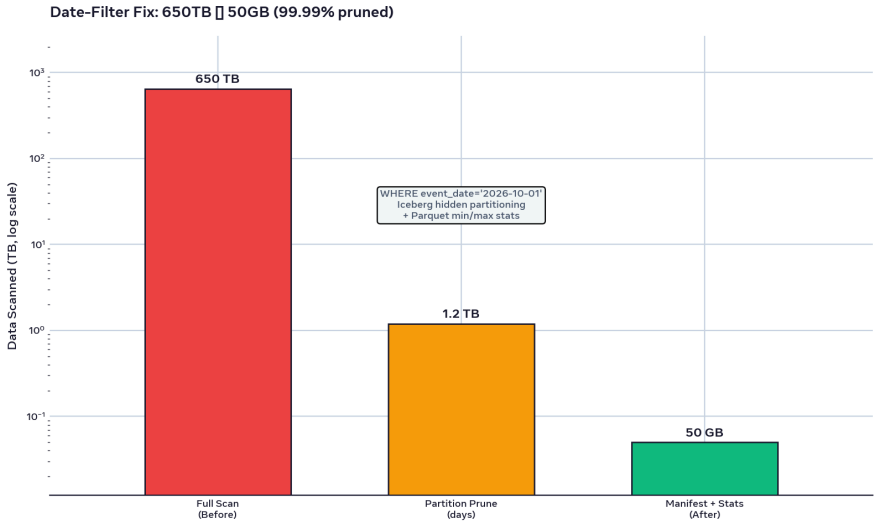
1 **Partition pruning**
Iceberg hidden partitioning days(event_time) • 1 day out of 730
650 → 1.2TB

2 **Manifest + column pruning**
Parquet min/max for user_id, event_time • file & column stats
1.2TB → 50GB

3 **Dashboard cache on Gold**
Gold aggregates in StarRocks, not Silver. 3-5 sec Athena/Trino ad-hoc.
3-5 sec

SQL: PARTITIONED BY days(event_time) SORTED BY user_id • TBLPROPERTIES write.format='parquet' codec='zstd'

DATE-FILTER FIX: 650TB → 50GB (LOG SCALE) ✓ PRUNED



Serving: dashboards & training

ANALYTICS • 15 MIN LAG

Gold tables in StarRocks / ClickHouse / Pinot

Dashboards read **gold.daily_kpi, funnel_hourly** — 512MB Parquet → dbt every 15 min. Sub-second queries, decoupled compute/storage.

```
-- StarRocks on s3://lake/gold/
SELECT date, COUNT(*) FROM gold.daily_kpi
WHERE date = CURRENT_DATE - 1 • p95 120ms
```

Trino on Silver for ad-hoc | Manifest pruning 99.9% | No full scan

15m	<1s	$ / TB
DBT FRESHNESS	DASHBOARD P95	COMPUTE SEPARATE

TRAINING • T+1 BATCH

Iceberg time travel → Feast offline → Spark

Training never hits online. Reads T-1 snapshot deterministically, exports to offline store. No impact on Scylla/StarRocks.

```
SELECT * FROM silver.events
FOR TIMESTAMP AS OF '2026-10-12'
-- → s3://features/offline/dt=2026-10-12/
```

Snapshot isolation | Replay any day | Zero skew

T-1	Feast	Spark
ICEBERG SNAPSHOT	ONLINE/OFFLINE CONSISTENCY	TRAINING CLUSTER

API layer: GraphQL federated over Scylla (online p95 10ms) + StarRocks (offline aggregates)

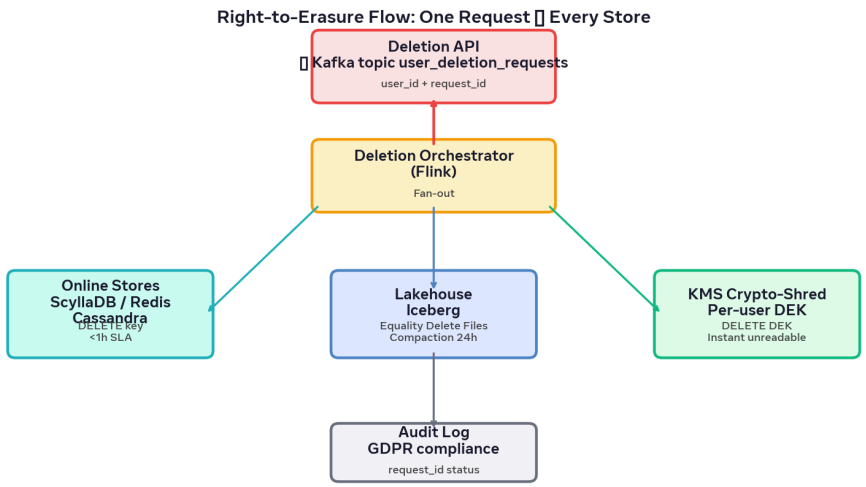
personalization → Scylla, analytics → StarRocks, feature flag routes by latency SLO • cache gold, not silver

S3 $15k/mo 650TB
vs Kafka $260k/mo
StarRocks computes separate

Erasure is a stream, not a script

TOPIC: USER_DELETION_REQUESTS

ONE REQUEST → EVERY STORE





Crypto-shred • DEK per user
KEK in KMS • field-level PII bytes = instant unreadable when DEK deleted. Tombstone not enough.

API → Kafka **user_deletion_requests** {user_id, request_id, requested_at}
Orchestrator (Flink) fans out • Audit log gdpr_audit(request_id, stores_purged)

1 ONLINE STORES • <1H SLA
DELETE Scylla / Redis / Cassandra keys
Scylla user_activity PK user_id, purchases_by_user

2 LAKEHOUSE • EQUALITY DELETES
Iceberg delete files → merged at read
Nightly rewrite_data_files physical removal • manifest pruning intact

3 KMS • CRYPTO-SHRED INSTANTANEOUS
DELETE KMS DEK per user
PII bytes already encrypted • key deletion = unreadable in 7d+ tiered S3 too

<1h	<24h	7d
ONLINE	LAKE COMPACTION	KAFKA RETENTION
		SOLVES

Why this stack?

Every choice optimizes for one clock while keeping erasure auditable. Iceberg wins for GDPR because equality deletes and partition evolution are first-class.

STREAMING • TRANSPORT

Kafka + Flink

Exactly-once checkpoints to S3, SQL filter, 72 partitions @ 500 eps. Low latency, backpressure aware.

> Kinesis: vendor lock, no RocksDB state. > Spark streaming: 2-min > Flink 10-sec.

LAKE • TRUTH

S3 + Iceberg Silver

Hidden partitioning days(event_time), Z-order user_id, equality deletes, time travel.

> Delta/Hudi: delete-by-rewrite, no native equality. Iceberg = GDPR compliant, partition evolution.

ONLINE • SPEED

ScyllaDB / DynamoDB

Scylla p95 10ms at 600M writes/day, 5× cheaper than Redis. DynamoDB if managed is priority.

> Redis: cost explodes at 1.2 TB/day. Scylla = LSM, wide rows, TTL 2y.

OLAP • 15M DASHBOARDS

StarRocks / ClickHouse / Pinot

StarRocks = MySQL compat + separate compute, 12 min freshness on Gold. ClickHouse for speed, Pinot for real-time.

Gold aggregates dbt every 15m, not Silver. Trino for ad-hoc on Silver.

FEATURES • CONSISTENCY

Feast online + offline

One definition for online Scylla and offline Iceberg T-1 snapshot. No skew training/serving.

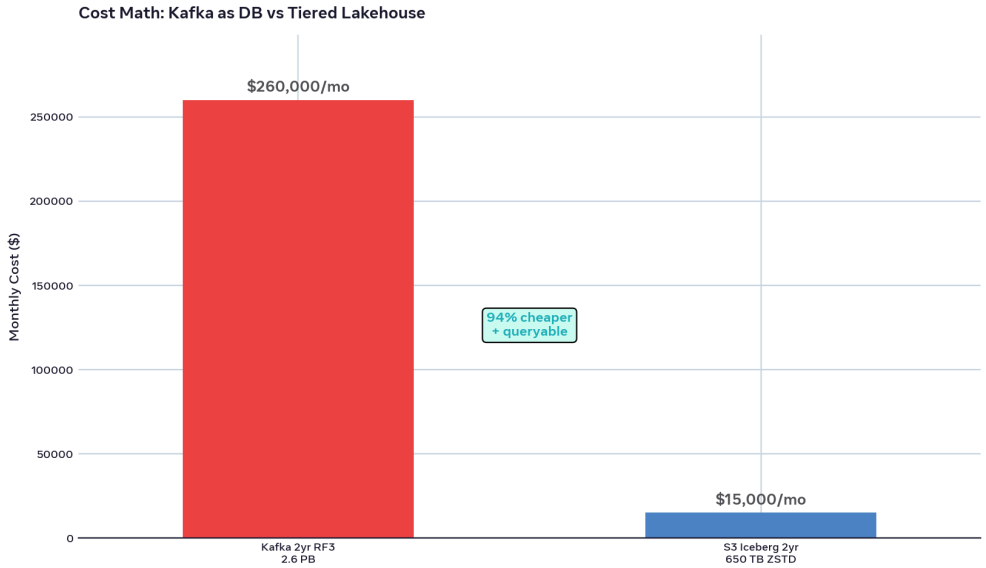
Iceberg FOR TIMESTAMP AS OF T-1 → offline store s3://features/offline/

✓ **Kappa for transport + Lakehouse for truth + Online for speed.** One event lands in Scylla in 2 sec, StarRocks in 12 min, T-1 snapshot tomorrow — without ever scanning 2 years.

Cost, monitoring, failure modes

S3 650 TB $15k/mo vs Kafka 2.6 PB $260k/mo

94% CHEAPER + QUERYABLE



$260,000/mo

KAFKA 2YR RF3 2.6 PB • $100/TB • NO PRUNING, NO SQL

$15,000/mo

S3 ICEBERG 650 TB ZSTD • $23/TB • DAYS() PRUNING + STATS

SMALL FILES SOLVED

Flink 60s rolling → 128 MB, Spark rewrite 512 MB

Target 512 MB Parquet reduces manifest from 50k → 1.3k files/day. Auto compact every 15 min.

LAG MONITORING • 4 CLOCKS

Flink backpressure, consumer lag, snapshot age, p95

```
consumer_lag{topic=events} < 2m
iceberg_snapshot_age < 15m
scylla_p95{tbl=user_activity} < 15ms
```

REPLAY FROM ANY DATE

Reprocess Bronze → Silver via Iceberg time travel

FOR TIMESTAMP AS OF '2026-10-01' → rewrites Silver slice. No replay through Kafka tiered log.

One log, four SLAs, zero full scans

We built **7-day Kafka + Tiered S3 + Flink speed layer + Iceberg lakehouse + StarRocks serving + deletion stream**.
Transport stays hot, truth lives in S3, speed lives in Scylla.

PERSONALIZATION

<3s to Scylla

p95 10ms, last 20 events

FRAUD

<2min filtered

30M purchase/day, 5-min window

ANALYTICS

15min, 50GB not 650TB

days(event_time) pruning

GDPR

Erasure as stream

equality deletes + KMS shred

NEXT • POC

Replay 1 day Bronze → Silver, benchmark partition pruning Trino 3-5 sec, load test Scylla 35K rps, write first **equality delete file**, validate KMS per-user DEK

Approve stack

Start with Bronze/Silver + personalization path → fraud filter → Gold aggregates 15m